Test Case / Original Request

Run the actual k6 test for PTC-007, track monitoring data, check the target’s security surface, and create a PDF summarizing the analysis with links to the generated artifacts.

REAL Execution Outcome

Execution status: COMPLETED
Runner status: completed
Outcome details: REAL k6 execution completed. Threshold breaches, if any, are retained as performance findings.
Thresholds passed: True

Executive Summary

Application: Unknown
Scenario: Shopping Cart Summary API — Load Check
Job ID: JOB-0d4725fae0bf435e8ef4a67b624f5b47
Overall Result: WARNING

Performance testing completed successfully: no failed requests and the average response time was within a reasonable range, though a high max response time created a warning due to outlier latency spikes. Infrastructure monitoring found load generator host and memory within healthy ranges. Security assessment was limited to configuration review with no controls tested, resulting in a warning—no runtime controls or live security checks were executed. Review the warning on response time spikes and consider augmenting security coverage in future runs.

Execution Profile and Artifacts

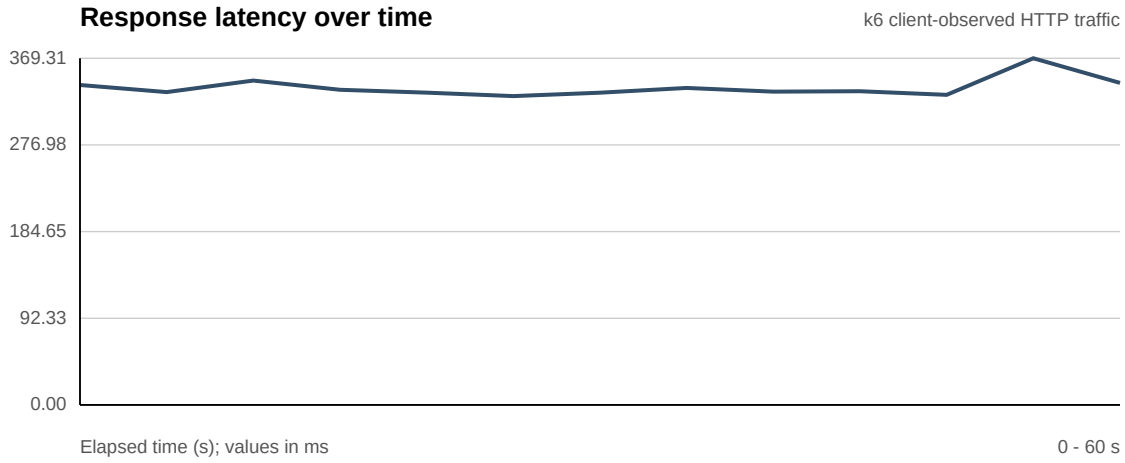
Execution mode: REAL
Workload profile:
- tool: k6
- target_url: https://dummyjson.com
- endpoint: /carts/1
- method: GET
- users: 60
- max_rps: 10
- duration_seconds: 60
- ramp_up_seconds: 5
- expected_status_code: 200
- p95_threshold_ms: 5000
Artifact directory: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47
Artifact references:
- aligned_time_series: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_aligned_time_series.csv
- generated_script: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check.js
- k6_summary: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_summary.json
- measurement_metadata: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_measurement_metadata.json

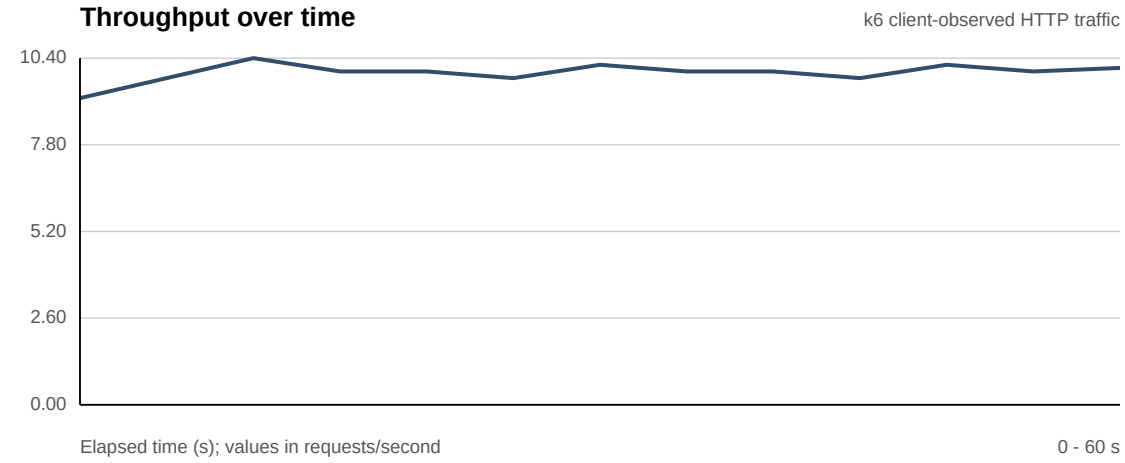
- monitoring_samples: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_runtime_samples.csv
- Native k6 HTML report
- Native report + raw results ZIP
- normalized_metrics: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_normalized_metrics.json
- Raw k6 results (JSON)
- runner_log: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_runner.log
- time_series: src/generated_scripts/k6 native reports/JOB-0d4725fae0bf435e8ef4a67b624f5b47/k6_PTC_007_Shopping_Cart_Summary_API_Load_Check_time_series.csv

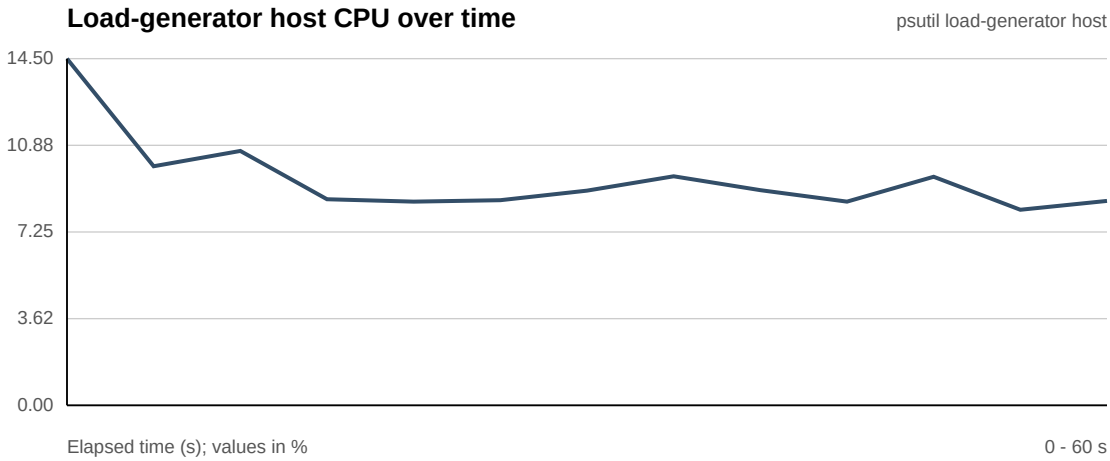
The interactive native HTML dashboard is linked as a separate artifact and is not embedded in this PDF.

Time-Series Charts (REAL Measurements)

Charts show measured values in five-second aligned buckets. Host CPU is from the load-generator machine, not the target.







Agent Activity Summary

Agent	*Summary*
Test Execution	Load test executed on /carts/1 (Unknown - Shopping Cart Summary API — Load Check). Results: 632 total requests, 632 successful, 0 failed. Avg response time: 338.20980537974714 ms; 95th percentile: 433.93129999999996 ms; Max: 899.475 ms. Throughput: 9.967266518950305 TPS; Error rate: 0.0%.
Monitoring	CPU usage: 9.41% average, 21.0% peak; memory: 67.81% average, 69.7% peak; DB connections: not collected, TPS: 9.967266518950305, Response time: 338.20980537974714 ms. Data source and scope: REAL: psutil load-generator host samples and k6 client-observed metrics. Measurement provenance: cpu_usage=psutil (load_generator_host); cpu_peak_usage=psutil (load_generator_host); memory_usage=psutil (load_generator_host); memory_peak_usage=psutil (load_generator_host); response_time=k6 (load_generator_observed_http_traffic); tps=k6 (load_generator_observed_http_traffic); db_connections=not_collected (No target-specific database telemetry provider is configured.). No alerts triggered.
Security Validation	Status: WARNING. Security score: N/A. Live security check: not requested. Issue: Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness. Issue: No security control was selected from the objective; no control was assessed. Control coverage: {'requested_count': 0, 'assessed_count': 0, 'not_assessed_count': 0} Assessment limitations: - No security control was selected from the objective; no control was assessed. - No runtime security checks were performed; results describe configuration declarations only.

Agent	*Summary*
Analysis	Performance testing completed successfully: no failed requests and the average response time was within a reasonable range, though a high max response time created a warning due to outlier latency spikes. Infrastructure monitoring found load generator host and memory within healthy ranges. Security assessment was limited to configuration review with no controls tested, resulting in a warning—no runtime controls or live security checks were executed. Review the warning on response time spikes and consider augmenting security coverage in future runs.
Reporting	A PDF report was generated, including all findings, recommendations, and an executive summary.

Per-API Findings (Test Execution)

- /carts/1 [WARNING / medium]
avg=338.20980537974714ms p95=433.93129999999996ms max=899.475ms throughput=9.967266518950305 req/s error_rate=0.0% http_status=200
* Max response time (899.475ms) is 2.1x p95 (433.93129999999996ms) -- outlier spikes present.
history: No comparable historical context was available for this API and scenario (RAG module unavailable); analysis is based only on detected threshold breach.
recommendation: Outlier spike in max response time warrants investigation. Since context is unavailable, start by checking application and infrastructure logs around the spike time; compare to runbook thresholds if documented. If repeated in future runs, create an incident and establish a baseline from new data.

Infrastructure Health (Monitoring)

Status: PASS (monitoring_agent reported: OK)
CPU usage: 9.41% Memory usage: 67.81% DB connections: None TPS: 9.967266518950305
Response time (monitoring): 338.20980537974714ms

Security Findings (Security Agent)

Status: WARNING Security score: N/A
Security result job ID: JOB-0d4725fae0bf435e8ef4a67b624f5b47.
Live security check: not requested.
Issues:
- Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness.
- No security control was selected from the objective; no control was assessed.
Control coverage: {'requested_count': 0, 'assessed_count': 0, 'not_assessed_count': 0}
Assessment limitations:
- No security control was selected from the objective; no control was assessed.
- No runtime security checks were performed; results describe configuration declarations only.

Recommended Next Steps

- /carts/1: Outlier spike in max response time warrants investigation. Since context is unavailable, start by checking application and infrastructure logs around the spike time; compare to runbook thresholds if

documented. If repeated in future runs, create an incident and establish a baseline from new data.
- Security: address the issues/recommendations listed above before next deployment.